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Featured Article

Incidence of dementia and subtypes: A cohort study in four regions in China

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Abstract

Background: There is a dearth of literature on the incidence of dementia in China.
Methods: Using a stratified, multistage, cluster-sampling method, 16,921 nondemented participants ≥55 years were recruited from four regional centers in China in 1997 and followed up to 4.5 years. Cases were identified through a three-step protocol, according to standardized criteria for dementia, Alzheimer's dementia (AD), and vascular dementia (VaD).
Results: The crude incidence in persons ≥65 years was 12.1/1000 person-years for dementia, 8.2/1000 person-years for AD, and 3.1/1000 person-years for VaD. After adjusting for sociodemographic factors, older age and lower education were associated with a higher risk of incident dementia, AD and VaD; regional difference was associated mainly with incidence of VaD (north vs. south: hazard ratio [HR] = 3.59); modestly with AD (east vs. west: HR = 1.55).
Conclusions: The incidence of dementia in Chinese population is comparable with that in Europe and United States.
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Keywords: Dementia; Alzheimer's disease; Vascular dementia; Incidence; Risk factors; Prospective cohort study; Epidemiology; China

1. Introduction

Dementia—which is characterized by deterioration in cognition, function, and behavior—places an enormous burden on the society. In China, the prevalence of Alzheimer's disease (AD) and vascular dementia (VaD) are comparable with those in Western countries, with positive associations for older age, lower education, and particular regions [1,2]. The incidence rate (that is, the rate of number of new cases of disease in a population over a period of time) has been found to be critical to assessing the disease etiology and is associated with potential risk factors.

Data on the incidence rate of dementia in China are scarce. According to studies involving mostly Caucasians, the rate of dementia ranges from 6.8 to 20.3 cases/1000 person-years (p-yrs) in individuals ≥65 years [3–8]. The average annual incidence of dementia and its subtypes:

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AD, VaD, and dementia by other causes has been reported to be 0.90%, 0.74%, 0.33%, and 0.08%, respectively [9,10]. Prince et al. reported incidence rates of 24.0/1000 p-yrs using their 10/66 instruments for case ascertainment, that is one-stage comprehensive diagnostic procedures including four scales: CSI "D," animal naming verbal fluency task, 10-word list learning task and geriatric mental state–automated geriatric examination for computer-assisted diagnosis (GMS-AGECAT), and 15.1/1000 p-yrs as assessed by DSM-IV criteria, based on a sample from urban and rural areas of Beijing [11–13]. Chen et al. [14] reported an incidence rate of 15.7/1000 p-yrs using GMS-AGECAT diagnosis based on a sample from Anhui Province. Although great strides have been made to establish the incidence rate of dementia in China, previous estimates have been extrapolated from data involving regional or limited size samples. China encompasses a very large geographical area and a widely diversified population comprising many different ethnic, cultural, and socioeconomic aspects. Thus, the local studies that have been published to date are limited in reflecting the overall epidemiologic situation of dementia in China. Because of major methodological differences between the local studies that have been published to date, it has not been feasible to derive a cross-country estimate of dementia incidence by integrating these studies. With the accelerated aging trend of the Chinese population (the latest statistics indicate that there are approximately 119 million people aged ≥65 years) [15,16], representative age-specific, sex-specific incidence rates of dementia in our country, as well as statistical comparisons of urban-rural, northern-southern, and eastern-western differences, are needed to target interventions and develop preventive strategies. Hence, to better characterize the overall epidemiologic profile of dementia in China, we conducted a population-based, prospective, cohort study in four geographically representative major regional centers in China, using extensive case ascertainment procedures.

2. Methods

2.1. Study areas

Similar to a study on the prevalence of dementia that we conducted in 1997 [1], we selected Beijing (northeast), Xi'an (northwest), Shanghai (southeast), and Chengdu (southwest) as the regions to perform our study, as we consider these regions to be representative of the broad spectrum of environmental, natural resource-related, economic, diet, cultural, and lifestyle variations across China.

2.2. Sample

Using a stratified, multistage, cluster-sampling methodology, population-based prevalence surveys were conducted on all people aged ≥55 years in the four aforementioned regions of China, which, as a subsample of the 1997 study [1], was adjusted and randomly selected at baseline based on sample size calculation with considering cost efficiency. The comparison of baseline demographic characteristics between samples entering the incidence study and that not entering was not substantially different.

At baseline, we recruited all eligible residents (n = 17,499) from 29 communities in 12 urban districts and 57 communities in 15 rural counties. In total, 16,921 nondemented participants entered the longitudinal study at baseline, and 578 residents were excluded due to dementia. Our initial, small-scale, pilot follow-up visit, including 1055 participants from 21 urban and 44 rural communities, was conducted in 1999. Subsequently, between years 2000 and 2002, we conducted a second, large-scale follow-up visit comprising the full sample from 29 urban and 57 rural communities (Fig. 1). To ensure a high follow-up rate, we revisited the participant's household up to four occasions. If the participants had moved, we sought their vital status and current residence from contact information recorded at baseline, and sought to reinterview them, even though they may have moved outside the catchment area.

2.3. Clinical interviewer training

We trained 63 interviewers consisting of neurologists, psychiatrists, and medical students from university hospitals. The regional supervisors received 2 weeks of training and community practice in Beijing. All field workers completed training in their region, including pilot interviews in nonsampled communities. At the completion of training, the interrater reliability of the cognitive tests was assessed using ratings of a videotaped model interview. During fieldwork, the principal investigator and supervisors monitored interviews on site. Retraining was undertaken every 6 months.

2.4. Informed consent

We obtained multilevel ethics approvals from institutional review boards and community leaders. Participants were informed about the study at local civic meetings and through public announcements. All eligible residents were invited to participate in a voluntary basis and gave oral consent.

2.5. Dementia assessments and diagnosis

All participants were interviewed for cognitive impairment in their residence or local hospital by trained interviewers. The full details of these assessment and diagnostic procedures have been previously described [1]. Briefly, we used multiphase assessments for case ascertainment. In phase I (dementia screening), a door-to-door screening survey was conducted, during which the Chinese version of the mini mental state examination (C-MMSE) [17] and Chinese version of activities of daily life (CADL) [18,19] were administered. A demographic questionnaire was also administered to gather the participant's personal

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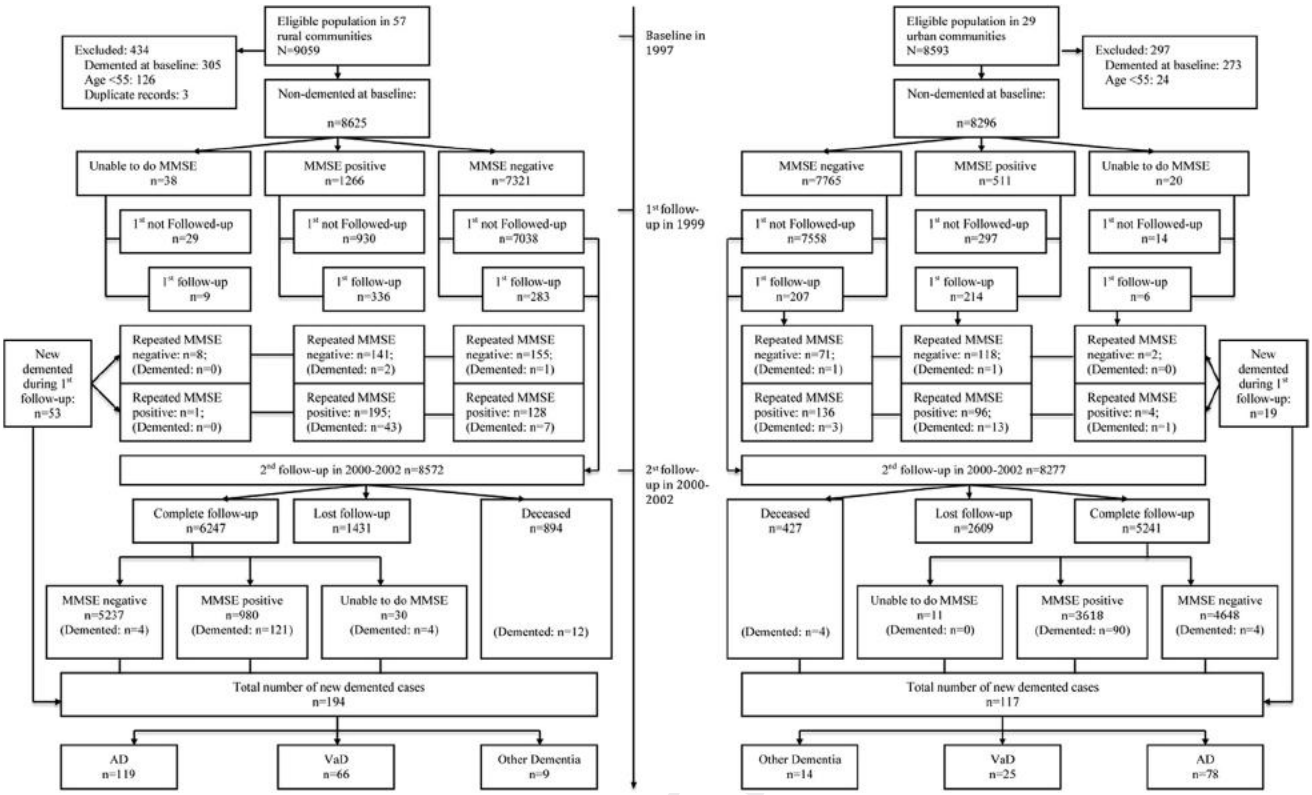

Fig. 1. Flow chart of the study. Abbreviations: MMSE, mini mental state examination; VaD, vascular dementia; AD, Alzheimer's dementia.

and medical history, as well as family history for dementia, and a brief physical and neurologic examination was performed. The cutoffs of C-MMSE were ≤19 points for illiterates, ≤22 for ≤6 years of education, and ≤26 for ≥7 years of education [17]. The CADL scale consists of 20 items including 8 instrumental items and 12 basic items. Each item was scored ranging from 1 to 4. Total scores for CADL range from 20 to 80. The cutoff point for abnormal CADL is ≥26. In phase II (clinical diagnostic workup), all persons scoring below the C-MMSE cutoff for their educational level and those unable to complete screening were interviewed to obtain detailed medical histories, standard neurologic examinations, and neuropsychological tests [9], as well as the Hamilton rating scale for depression [20], and the Hachinski ischemia scale [21]. Proxy (caregiver) interviews and a Chinese care-giver questionnaire [22] were completed when physical impairment or illness precluded direct examination. In phase III (6-month diagnostic confirmation), a preliminary diagnosis of dementia was confirmed by repeating all screening tests and diagnostic examinations 6 months later for participants in every phase II who could be traced (67.89% were re-tested). After the phase III examinations, a panel of senior clinicians at each site confirmed the final diagnosis on the basis of all available data according to the criteria of DSM-IV for dementia [23], the criteria of the National Institute of Neurological and Communicative Diseases and Stroke—Alzheimer Disease and Related Disorders Association [24] for probable or possible AD, and those of the National Institute of Neurological Disorders and Stroke—Association Internationale pour la Recherche et l'Enseignement en Neurosciences [25] for probable or possible VaD.

Participants identified as nondemented were followed and re-examined during the incidence wave. Same standard three-step procedures used for cognitive evaluation were repeated to determine the onset of dementia for nondeceased participants. For deceased participants, we recorded the date and the final diagnosis of their death and performed interviews with their next-of-kin and community leaders, to seek evidence of cognitive and functional decline suggestive of dementia onset before death. The date of dementia onset was estimated (and recorded as the mid-date of the year indicated) by retrospectively interviewing the incident cases and their relatives, and by reviewing the medical records, if available.

2.6. Statistical analysis

Incidence rates of dementia per 1000 p-yrs were calculated as the number of new cases divided by the number of person-years at risk. Person-years were calculated as the interval from baseline to the time of dementia onset, time of death, or to the time of follow-up assessment. Age- and sex-specific incidence rates by 5-year groups were calculated based on the baseline age. The 95% confidence intervals (CIs) were based on the Byar's distribution. The

Table 1
Baseline characteristics of participants from the four regional centers and those who completed or who were lost to follow-up

Characteristics	Beijing (N.E. Region)	Xian (N.W region)	Shanghai (S.E. Region)	Chengdu (S.W. Region)	Rural (All regions)	Urban (All regions)	Total traced (All regions)	Lost to follow-up (All regions)
n	3602	2202	3835	3242	7194	5687	12,881	4040
Age at baseline, n (%)								
55–59	744 (20.7)	500 (22.7)	454 (11.8)	719 (22.2)	1651 (23.0)	766 (13.5)	2417 (18.8)	773 (19.1)
60–64	982 (27.3)	574 (26.1)	748 (19.5)	686 (21.2)	1635 (22.7)	1355 (23.8)	2990 (23.2)	925 (22.9)
65–69	817 (22.7)	453 (20.6)	927 (24.2)	740 (22.8)	1500 (20.9)	1437 (25.3)	2937 (22.8)	883 (21.9)
70–74	528 (14.7)	325 (14.8)	787 (20.5)	545 (16.8)	1157 (16.1)	1028 (18.1)	2185 (17.0)	724 (17.9)
75–79	311 (8.6)	230 (10.5)	516 (13.5)	307 (9.5)	746 (10.4)	618 (10.9)	1364 (10.6)	407 (10.1)
80–84	151 (4.2)	91 (4.1)	259 (6.8)	155 (4.8)	342 (4.8)	314 (5.5)	656 (5.1)	218 (5.4)
≥85	69 (1.9)	29 (1.3)	144 (3.8)	90 (2.8)	163 (2.3)	169 (3.0)	332 (2.6)	110 (2.8)
Female sex, n (%)	1921 (53.3)	1234 (56.0)	2110 (55.0)	1617 (49.9)	3751 (52.1)	3131 (55.1)	6882 (53.4)	2258 (55.9)
Education, y, n (%)								
<1	1344 (37.3)	1117 (50.7)	1101 (28.7)	1361 (42.0)	3694 (51.4)	1229 (21.6)	4923 (38.2)	1264 (31.3)
1–6	1049 (29.1)	674 (30.6)	1243 (32.4)	1532 (47.3)	2771 (38.5)	1727 (30.4)	4498 (34.9)	1300 (32.2)
7–12	598 (16.6)	352 (16.0)	1158 (30.2)	322 (9.9)	679 (9.4)	1751 (30.8)	2430 (18.9)	1005 (24.9)
≥12	611 (17.0)	59 (2.7)	333 (8.7)	27 (0.1)	50 (0.7)	980 (17.2)	1030 (8.0)	471 (11.7)
Residence, n (%)								
Urban	1765 (49.0)	711 (32.3)	2595 (67.7)	616 (19.0)	—	—	5687 (44.1)	2609 (64.6)
Rural	1837 (51.0)	1491 (67.7)	1240 (32.3)	2626 (81.0)	—	—	7194 (55.9)	1431 (35.4)
Baseline MMSE, mean (SD)	26.1 (8.7)	25.6 (6.0)	27.3 (3.1)	24.3 (4.9)	24.6 (6.8)	27.5 (4.7)	25.9 (6.2)	26.6 (4.9)

Abbreviations: MMSE, mini mental state examination; SD, standard deviation.
NOTE. Beijing and Xi'an in North; Beijing and Shanghai in East; Shanghai and Chengdu in South; and Xi'an and Chengdu in West.

age-standardized overall rates were calculated by direct method using data from the 2010 Chinese population census [15] or the 2010 United States census [26] as standardization weights. A Cox proportional hazard multivariate model was used to identify the effect of demographic variables in the risk of developing dementia. Age was entered in the model as a categorical variable in 5-year intervals. Education was categorized as "illiterate," "primary school," "middle school," and "college and above." Sex, residence status, and geographical regions were entered as dichotomous variables; that is, females versus males, rural areas versus urban areas, northern versus southern region, and eastern versus western region. In model 1, no covariates were adjusted for each independent variable. In model 2, age was adjusted as a covariate for every other independent variable. In model 3, age, gender, education, residence status, and region 1 (northern versus southern region) were entered as independent variables. In model 4, region 2 (Eastern versus Western region) with all other variables were entered as independent variables. Interaction of age and education was tested and removed from the model due to a lack of statistical significance. All the analysis was performed using SAS 9.3 software (SAS Institute Inc, 2011).

3. Results

3.1. Baseline characteristics

Of the 16,921 participants, 12,881 participants could be traced, and they comprised 11,560 alive and 1321 deceased participants. A total of 4040 participants (23.9%) were lost to follow-up because of refusal to participate in the study (2.4%) or our inability to trace them (21.5%). The median follow-up time is 3.1 (interquartile, 2.9–3.5) years. The baseline characteristics of age and gender distribution between the traced and the lost to follow-up group were similar, except that in the lost to follow-up group, the percentage of participants living in urban areas and being better educated was higher than those in the traced group. Further comparisons of demographic characteristics of each regional center, rural, and urban areas are shown in Table 1.

3.2. Crude IR and specific rates by age, gender, residence status, and region Q2

In total, 311 new cases of dementia were identified and included in the analysis with a total 39,893 p-yrs of follow-up. The crude incidence in participants ≥55 years was 7.8/1000 p-yrs (95% CI, 7.0–8.7/1000 p-yrs) for dementia, 4.9/1000 p-yrs (95% CI, 4.3–5.7/1000 p-yrs) for AD, and 2.3/1000 p-yrs (95% CI, 1.9–2.8/1000 p-yrs) for VaD. The crude incidence in participants ≥65 years increased to 12.1/1000 p-yrs (95% CI, 10.8–13.7/1000 p-yrs) for dementia, 8.2/1000 p-yrs (95% CI, 7.0–9.4/1000 p-yrs) for AD, and 3.1/1000 p-yrs (95% CI, 2.5–3.9/1000 p-yrs) for VaD. The specific rates by age, gender, education levels, residence status, and geographical regions were shown in Table 2. The IR increased progressively with age in 5-year intervals, reaching 64.4/1000 p-yrs for individuals aged >85 years. The rates in females were higher than males, especially for those >70s-years-old, but reached similar levels to males after 85 years. In rural areas, the rates were 8.7/1000 p-yrs for

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Table 2
Incidence rates for all dementia, AD and VaD

	All dementia		AD		VaD	
Characteristics	New cases	IR (95% CI)	New cases	IR (95% CI)	New cases	IR (95% CI)
Crude (age ≥55)	311	7.80 (6.97–8.70)	197	4.94 (4.28–5.66)	91	2.28 (1.85–2.79)
Crude (age ≥65)	268	12.14 (10.75–13.66)	180	8.15 (7.03–9.41)	69	3.13 (2.45–3.93)
Age, y						
55–59	15	1.87 (1.09–3.01)	6	0.75 (0.31–1.54)	7	0.87 (0.39–1.71)
60–64	28	2.86 (1.94–4.07)	11	1.12 (0.60–1.94)	15	1.53 (0.89–2.46)
65–69	33	3.56 (2.49–4.93)	15	1.62 (0.94–2.60)	15	1.62 (0.94–2.60)
70–74	63	9.63 (7.47–12.24)	39	5.96 (4.30–8.06)	18	2.75 (1.69–4.26)
75–79	75	19.69 (15.60-24.54)	49	12.86 (9.63-16.86)	21	5.51 (3.51-8.27)
80–84	50	29.19 (21.91–38.15)	40	23.35 (16.93–31.46)	9	5.25 (2.59–9.59)
≥85	47	64.30 (47.82–84.72)	37	50.62 (36.21–68.98)	6	8.21 (3.41–16.92)
Gender						
Male	117	6.34 (5.27–7.57)	68	3.69 (2.89–4.64)	38	2.06 (1.48–2.80)
Female	194	9.05 (7.84–10.39)	129	6.02 (5.04–7.12)	53	2.47 (1.87-–3.21)
Education, y						
<1	193	12.87 (11.15–14.78)	124	8.27 (6.91–9.82)	56	3.73 (2.85–4.81)
1–6	86	6.20 (4.99–7.62)	54	3.89 (2.96–5.04)	27	1.95 (1.31–2.79)
7–12	22	2.90 (1.87–4.31)	12	1.58 (0.86–2.68)	7	0.92 (0.41–1.81)
≥12	10	2.90 (1.49–5.15)	7	2.03 (0.91–3.99)	1	0.29 (0.03–1.35)
Residence						
Urban	117	6.69 (5.56–7.99)	78	4.46 (3.55–5.53)	25	1.43 (0.95–2.08)
Rural	194	8.66 (7.50–9.94)	119	5.31 (4.42–6.33)	66	2.95 (2.30–3.72)
Region 1						
South	150	7.30 (6.20–8.54)	113	5.50 (4.55–6.58)	25	1.22 (0.81–1.77)
North	161	8.32 (7.11–9.69)	84	4.34 (3.49-5.35)	66	3.41 (2.66-4.31)
Region 2						
West	91	5.67 (4.60–6.93)	64	3.99 (3.10–5.06)	21	1.31 (0.83–1.96)
East	220	9.22 (8.06–10.50)	133	5.58 (4.69–6.58)	70	2.93 (2.31–3.68)

Abbreviations: AD, Alzheimer's dementia; VaD, vascular dementia; CI, confidence interval.

individuals ≥55-year-old and 14.0/1000 p-yrs for individuals ≥65 years and these rates were higher than the corresponding rates in urban areas. We also observed that rates were higher in northern areas 8.3/1000 p-yrs (95% CI, 7.1–9.7/1000 p-yrs) than in southern areas 7.3/1000p-yrs (95% CI, 6.2–8.5/1000 p-yrs.), as well as in eastern areas 9.2/1000 p-yrs (95% CI, 8.1–10.5/1000 p-yrs) than in western areas 5.7/1000 p-yrs (95% CI, 4.6–6.9/1000 p-yrs). The age-specific IR by gender, residence status, and regions are depicted in Fig. 2. We can observe that female and rural residents have higher IR, especially among those ≥75-years-old. Participants in northern areas and eastern areas have higher IR in the <85-year age groups but were similar for the ≥85-year age group. For subtypes of dementia, the incidence rate of AD was higher and increased much more dramatically with age than that of VaD.

3.3. Cox regression analysis for IR and age, gender, education, residence status, and regions

The association between IR and age, gender, education, residence status (rural versus urban), and regions (north versus south, east versus west) using Cox regression analysis are shown in Table 3. In model 1, increasing age, female gender, lower education, and living in rural, northern, and eastern areas were positively associated with incidence of dementia. In model 2, after controlling for age, the association observed in model 1 did not change. In model 3, with all covariates of age, gender, education, residence status, and region (north versus south) in the model, gender and residence status did not show significant association with incident dementia. Increasing age, lower education, and living in northern areas still showed positive associations. In model 4, with all covariates of age, gender, education, residence status, and region (east versus west) in the model, living in eastern areas was positively associated with incident dementia, with no change of association for age and education and only slightly positive association for living in rural areas. Further analysis for subtypes of dementia, besides the invariable association of age and education for both types, we found, for AD, no difference between north and south and only mild difference between east and west; for VaD, stronger association was observed for north versus south and east versus west.

3.4. Comparison of the present study in China with international studies

As shown in Fig. 3, the age-specific incidence rates of dementia in China were similar to those reported in Europe and United States and even showed a more rapid increase after age 85 year. Further analysis (by applying direct standardization with the age distribution of the Chinese and the US

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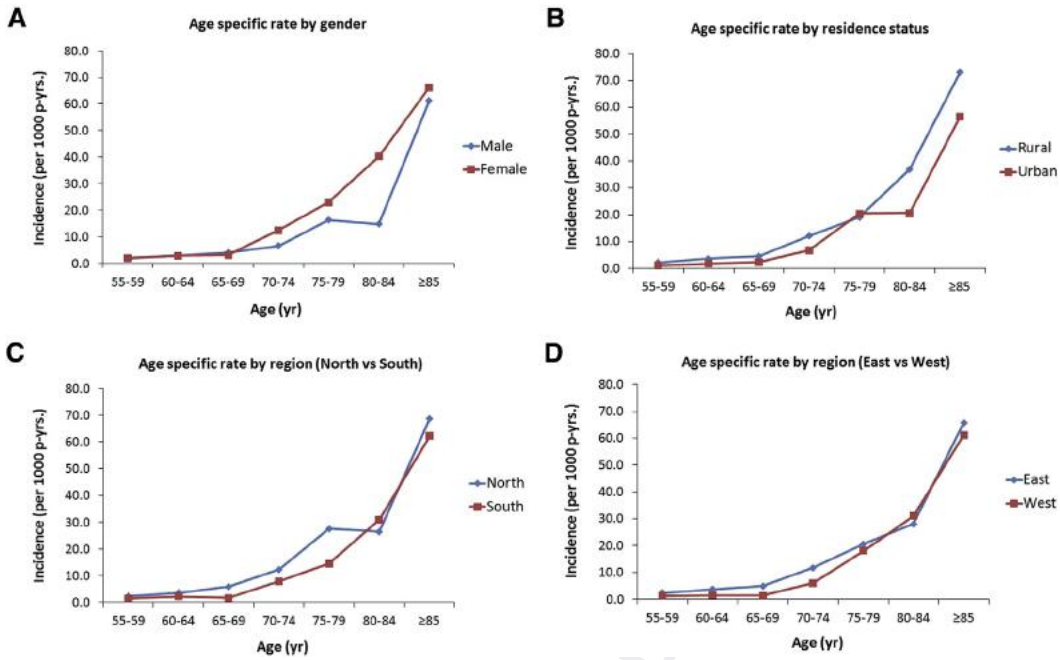

Fig. 2. Age-specific IR by (A) gender, (B) residence status, and regions ([C] North versus South and [D] East versus West).

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populations in year 2010 as the standards for participants ≥65 years) showed that the age-standardized overall IRs in the present study are 15.5/1000 p-yrs and 20.3/1000 p-yrs, respectively, which are comparable with the standardized overall rates of the EURODEM study [6] and the Rotterdam study [7] in Europe, the ACT study [3], the Rochester study [4], and the MoVIES study [5] in the United States (Table 4).

4. Discussion

This study is, to our knowledge, the largest multiregional prospective cohort dementia incidence study in China. We found that the age-specific incidence rates of dementia in the Chinese population increases exponentially with age and is comparable with those reported in developed countries [3–5,7,8,27]. Adjusting for all sociodemographic factors, older age, and lower education were associated with higher risk of incident dementia; regional difference was associated mainly with incidence of VaD, modestly with AD. Living in rural areas was associated with VaD but not AD. Although being female was associated with a higher incidence of dementia, no statistically significant association was observed after controlling for age and education.

Several factors may contribute to the high incidence of dementia in China. In our study, aging was found to be the most important risk factor; the incidence almost doubles with every 10-year increase in age, especially for AD. Given that the current life expectancy in China is comparable with that of developed countries, over time, the number of newly diagnosed cases is expected to reach a similar rate to that in western countries. Additionally, as shown in our study, the level of education in China is relatively low; less than one-third of people have attained middle school education or higher, especially in the rural areas, where only 10% of people have middle school qualifications or higher. Lower education levels have previously been found to be a risk factor for incident dementia [3,5,6,28] and may partly account for the high incidence in China, especially among the oldest age group, which received even less education during schooling age because of the war.

Disparities between geographical regions in incidence rates exist, although are not substantially different for AD, that were significantly observed for VaD. Higher prevalence of VaD has been reported in the north than in the south [2]. Higher prevalences of hypertension, diabetes, hyperlipidemia, and stroke were reported in the north and these were linked with multiple lifestyle factors, such as, physical inactivity, habitual drinking, and smoking [29–33]. The unoptimal control of vascular disease and exposure to risk factors is a potential risk for VaD [34], the higher incidence of dementia in the north may be related to a higher VaD incidence rate as shown by the study. The observed difference between east and west could be driven by the higher VaD in Beijing where it had high incidence of stroke [31]. Another possible explanation is that there might be more stroke survivors in the east because of more economically developed and better medical care service available there, although need to be approved by further better designed studies [35]. As observed in our study, only modestly elevated AD hazard ratio in eastern as compared with western regions of China. Longer life expectancy could be a possible reason for the higher incidence of AD in the east [35], while a statistical effect stemming from the large sample size can not be excluded as well.

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Table 3
Cox proportional hazards regression analysis for all dementia, AD and VaD

Characteristics	Model 1, no covariate adjusted* HR (95% CI)	Model 2, age covariate adjusted† HR (95% CI)	Model 3, full covariate adjusted‡ HR (95% CI)	Model 4, Full covariate adjusted§ HR (95% CI)
All dementia				
Age, y				
55–59	Reference*	—	Reference	Reference
60–64	1.51 (0.81–2.84)	—	1.43 (0.76–2.69)	1.37 (0.73–2.58)
65–69	1.85 (1.00–3.40)	—	1.76 (0.95–3.25)	1.61 (0.87–2.98)
70–74	4.88 (2.78–8.57)	—	4.57 (2.58–8.11)	4.04 (2.28–7.16)
75–79	9.70 (5.57–16.89)	—	9.01 (5.12–15.84)	7.88 (4.49–13.86)
80–84	14.02 (7.87–24.98)	—	12.48 (6.91–22.54)	10.66 (5.91–19.24)
≥85	28.71 (16.04–51.36)	—	25.87 (14.19–47.18)	21.19 (11.65–38.53)
Gender				
Male	Reference	Reference	Reference	Reference
Female	1.43 (1.14–1.80)	1.41 (1.12–1.77)	1.14 (0.88–1.48)	1.06 (0.82–1.38)
Education, y				
<1	Reference	Reference	Reference	Reference
1–6	0.48 (0.37–0.62)	0.65 (0.50–0.84)	0.75 (0.56–1.00)	0.69 (0.51–0.92)
7–12	0.22 (0.14–0.35)	0.33 (0.21–0.51)	0.40 (0.24–0.64)	0.34 (0.21–0.56)
≥12	0.24 (0.12–0.44)	0.39 (0.20–0.74)	0.43 (0.22–0.85)	0.40 (0.20–0.78)
Residence				
Urban	Reference	Reference	Reference	Reference
Rural	1.33 (1.05–1.67)	1.49 (1.19–1.88)	1.13 (0.88–1.46)	1.35 (1.04–1.76)
Region 1				
South	Reference	Reference	Reference	—
North	1.28 (1.02–1.60)	1.65 (1.32–2.07)	1.58 (1.26–1.99)	—
Region 2				
West	Reference	Reference	—	Reference
East	1.73 (1.35–2.21)	1.52 (1.19–1.95)	—	1.92 (1.49–2.48)
AD				
Age, y				
55–59	Reference		Reference	Reference
60–64	1.49 (0.55–4.02)		1.42 (0.52–3.84)	1.38 (0.51–3.75)
65–69	2.10 (0.81–5.40)		1.98 (0.76–5.13)	1.90 (0.73–4.92)
70–74	7.53 (3.19–17.80)		6.98 (2.92–16.64)	6.57 (2.76–15.65)
75–79	15.80 (6.77–36.88)		14.55 (6.17–34.32)	13.60 (5.77–32.07)
80–84	28.00 (11.87–66.04)		24.40 (10.18–58.47)	22.57 (9.43–54.05)
≥85	56.50 (23.84–133.93)		48.08 (19.88–16.30)	43.97 (18.22–106.09)
Gender				
Male	Reference	Reference	Reference	Reference
Female	1.64 (1.22–2.20)	1.59 (1.18–2.13)	1.32 (0.95–1.84)	1.27 (0.91–1.77)
Education, y				
<1	Reference	Reference	Reference	Reference
1–6	0.47 (0.34–0.65)	0.68 (0.49–0.93)	0.80 (0.55–1.15)	0.77 (0.54–1.12)
7–12	0.19 (0.11–0.34)	0.31 (0.17–0.56)	0.38 (0.20–0.71)	0.35 (0.18–0.66)
≥12	0.26 (0.12–0.55)	0.48 (0.23–1.04)	0.59 (0.26–1.34)	0.55 (0.24–1.26)
Residence				
Urban	Reference	Reference	Reference	Reference
Rural	1.23 (0.92–1.63)	1.41 (1.06–1.88)	1.12 (0.82–1.54)	1.28 (0.92–1.77)
Region 1				
South	Reference	Reference	Reference	
North	0.89 (0.67–1.19)	1.21 (0.91–1.62)	1.16 (0.87–1.55)	—
Region 2				
West	Reference	Reference	—	Reference
East	1.49 (1.11–2.01)	1.28 (0.95–1.73)	—	1.55 (1.13–2.13)
VaD				
Age, y				
55–59	Reference		Reference	Reference
60–64	1.74 (0.71–4.27)		1.71 (0.69–4.22)	1.56 (0.63–3.85)
65–69	1.81 (0.74–4.43)		1.87 (0.75–4.65)	1.55 (0.62–3.84)
70–74	3.01 (1.26–7.21)		3.13 (1.28–7.66)	2.39 (0.98–5.81)
75–79	6.17 (2.63–14.44)		6.24 (2.61–14.96)	4.75 (1.99–11.33)

(*Continued*)

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Table 3
Cox proportional hazards regression analysis for all dementia, AD and VaD (*Continued*)

Characteristics	Model 1, no covariate adjusted* HR (95% CI)	Model 2, age covariate adjusted† HR (95% CI)	Model 3, full covariate adjusted‡ HR (95% CI)	Model 4, Full covariate adjusted§ HR (95% CI)
80–84	6.09 (2.32–16.02)		6.12 (2.27–16.52)	4.45 (1.65–1.97)
≥85	8.00 (2.68–23.82)		9.19 (2.98–28.35)	5.80 (1.89–17.77)
Gender				
Male	Reference	Reference	Reference	Reference
Female	1.19 (0.79–1.80)	1.21 (0.80–1.83)	0.97 (0.61–1.56)	0.84 (0.53–1.34)
Education, y				
<1	Reference	Reference	Reference	Reference
1–6	0.51 (0.32–0.80)	0.60 (0.38–0.94)	0.76 (0.45–1.28)	0.60 (0.36–1.02)
7–12	0.24 (0.11–0.52)	0.30 (0.13–0.65)	0.44 (0.18–1.06)	0.32 (0.13–0.76)
≥12	0.08 (0.01–0.56)	0.10 (0.01–0.74)	0.13 (0.02–0.98)	0.11 (0.02–0.86)
Residence				
Urban	Reference	Reference	Reference	Reference
Rural	2.16 (1.36–3.42)	2.34 (1.48–3.71)	1.57 (0.95–2.59)	2.03 (1.22–3.38)
Region 1				
South	Reference	Reference	Reference	—
North	3.18 (2.01–5.04)	3.75 (2.36–5.96)	3.59 (2.25–5.73)	—
Region 2				
West	Reference	Reference	—	Reference
East	2.27 (1.41–3.67)	2.10 (1.30–3.39)	—	3.04 (1.86–4.97)

Abbreviations: AD, Alzheimer's dementia; VaD, vascular dementia; HR, hazard ratio; CI, confidence interval.
*Model 1 regression analysis for each independent variable adjusting for no additional covariates.
†Model 2 regression analysis for each independent variable adjusting for age group (seven levels) as a covariate.
‡Model 3 regression analysis for independent variables (and levels) include age group (7), gender (2), education (4), residence status (2), and region 1: north versus south (2).
§Model 4 regression analysis for independent variables (and levels) include age group (7), gender (2), education (4), residence status (2), and region 2: east versus west (2).

Previous studies in China were typically single-city studies or studies which enrolled fewer participants living in rural areas [10,12]. As shown in our study, the incidence rate in rural areas is higher than that in urban areas, which may explain the higher incidence rate in our study than in previous reports. Although lower vascular burden was reported among rural residents than the urban [29,30], rural residents tend to have lower education, unbalanced nutrition, poorer health care, and exposure to farm chemicals. These might cause them more vulnerable to poststroke cognitive impairment [36–38].

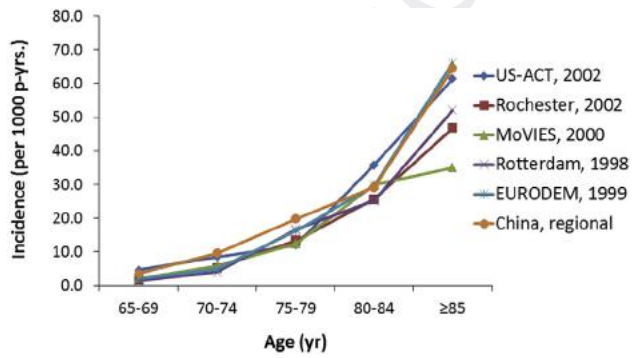

Fig. 3. Comparison of age-specific incidence rates of dementia across studies.

Our findings do not support the hypothesis that gender affects the incidence of dementia. Previous evidence of the association between gender and dementia has been inconclusive. Several studies have reported a higher incidence of dementia in women [6,7,39]. However, the ACT Cohort Study [3], the Rochester study [4], the MoVIES study [5], the Framingham Heart Study [8], and the Cardiovascular Health Study [40] provided no support for a positive association in women. Possible explanations for the higher incidence of dementia among women than men in our study is the longer life expectancies among women (which was reflected by the greater proportion of females than males in the older group) and the greater level of educational attainment among men than women. After adjusting for confounders, the incidence of dementia was not statistically significantly different between genders. In the Western world, dementia is generally more common in women than in men after age 85 years. In our study, dementia was more common among women aged 70–85 years. This discrepancy might be related to the educational level of females in China, which differs from those in western world. Females in this age group were born around 1910s–1930s when education for women was not easily accessible in China due to historical reasons.

Our study has its strengths and limitations. The previous studies investigating the incidence of dementia in China have been comparatively smaller and have involved

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Table 4
Age-standardized incidence rates of dementia in China versus studies in Western countries

Characteristics	China	US-ACT	Rochester	MoVIES	Rotterdam	EURODEM
Age, y						
65–69	3.6 (2.5–4.9)	4.7 (1.8–10.2)	1.7 (1.0–2.6)	2.1 (0.4–6.7)	1.4 (0.5–3.3)	2.0 (1.1–3.4)
70–74	9.6 (7.5–12.2)	8.4 (5.7–12.1)	5.2 (3.8–7.0)	5.8 (3.4–9.3)	3.9 (2.1–6.8)	4.9 (3.5–6.6)
75–79	19.7 (15.6–24.5)	12.0 (8.6–16.5)	13.3 (10.8–16.3)	12.4 (8.6–17.3)	16.7 (11.9–22.7)	16.2 (13.4–19.6)
80–84	29.2 (21.9–38.2)	35.8 (28.3–44.7)	25.6 (21.3–30.5)	29.8 (21.6–40.2)	25.2 (18.1–34.2)	29.7 (25.0–35.1)
≥85	64.3 (47.8–84.7)	61.3 (48.4–76.6)	46.7 (40.4–53.8)	35.0 (22.3–52.5)	52.0 (40.7–65.5)	66.1 (58.1–74.9)
All ages ≥65	12.1 (10.8–13.7)	20.3 (17.7–23.2)	13.6 (12.4–14.9)	13.6 (11.3–16.4)	14.8 (12.6–17.2)	18.4 (16.8–20.0)
Standardization*	15.5 (13.6–17.4)	14.3 (12.1–16.5)	10.6 (9.6–11.6)	10.4 (8.3–12.5)	11.1 (9.3–12.9)	12.9 (11.8–14.0)
Standardization†	20.3 (17.3–23.3)	19.0 (16.3–21.7)	14.2 (12.9–15.4)	13.3 (10.5–16.0)	15.1 (12.7–17.4)	18.0 (16.5–19.5)

*Standardized to China's population.
†Standardized to US population.

homogeneous geographical areas that yielded relatively fewer new dementia cases [10]. In comparison, our study is, to our knowledge, the first in China to be conducted in diverse regions and has a large sample, a high participation rate, and a relatively high follow-up rate. Standardized sampling, case-ascertainment, and follow-up procedures were used across sites. Moreover, the 6-month diagnostic confirmation served both to verify the progressive deterioration of cognitive function and to decrease diagnostic errors in preliminary assessments. As for the limitations of this study, first, attrition of the cohort may produce bias. Nonetheless, our study had a good response rate of 76%. In addition, the participants who were lost to follow-up did not differ significantly in baseline characteristics (including age, sex, and MMSE score) from those who remained in the study, except for educational and residential characteristics, where the proportion of higher educated and urban residents were higher in the lost-to-follow-up group. This observation may be related to migratory movements associated with new, large-scale building construction projects across many Chinese cities. Because urban residence and the attainment of higher education levels are associated with a lower incidence of dementia, our analysis is unlikely to be affected by this finding. Another limitation is relatively short follow-up time and potential inclusion of MCI subjects at baseline that might possibly be an overestimation of the rates.

Owing to China's decades-long birth control policy and large internal migration of working-age adults from rural to urban areas, fewer adults are available to provide continuing care to the millions of people with dementia, particularly in rural regions [41]. Clearly, anything that can be done to reduce the overall risk or to delay the time to dementia onset would be of huge benefit to both the public and the individual.

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RESEARCH IN CONTEXT

1. Systematic review: We searched PubMed in English and Wan-fang data in Chinese up to April, 2014, with no date limits set, with the terms "incidence," "dementia," "China," and "Chinese." We identified four population-based studies. All studies were completed in their local regions with different case ascertainment strategy and diagnosis criteria.
2. Interpretation: Our results advance in presenting more representative incidence data of dementia, Alzheimer's disease, and vascular dementia in Chinese population. The crude rates and the specific rates by age, gender, residence status, and geographical regions are the essential measures of disease frequency and distribution and indices of the disease impact on public health. The standardized rate enables the comparison between Chinese population and other ethnics.
3. Future directions: Further efforts in establishing longitudinal study with longer follow-up time and more sensitive neuropsychological instruments to recognize the disease in early stage would be important.

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